

Basics 36: Coupled Harmonics and Avoided Crossings

V - Plasma waves and Alfvén physics

Two-mode matrices, gap formation, harmonic mixing, positivity, and mode tracking

Textbook draft, version 1.0. Toroidal and stellarator geometry couples Fourier harmonics that would be independent in a uniform medium. The simplest two-harmonic model already explains avoided crossings, spectral gaps, mixed eigenvectors, and the numerical issues central to the project's reduced Module 4 benchmark. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1	Learning objectives	2
2	Prerequisites and reading strategy	2
3	Uncoupled two-mode model	2
4	Coupled matrix and eigenvalues	3
5	Gap at the uncoupled crossing	3
6	Harmonic mixing and mixing angle	3
7	Positivity-preserving coupling	4
8	From local matrix to radial differential operator	4
9	Mode tracking across scans	4
10	Physical meaning in toroidal plasmas	4
11	Computational laboratory	5
12	Summary	5

Symbols and acronyms used in this document	5
13 Exercises	5
14 References	6

1 Learning objectives

After completing this note, the reader should be able to:

- Solve a two-mode coupled eigenvalue problem analytically.
- Distinguish exact crossing from avoided crossing.
- Calculate gap width and harmonic mixing.
- Explain positivity constraints on a squared-frequency operator.
- Connect local coupling to a global radial eigenproblem.
- Track modes robustly across parameter scans.

2 Prerequisites and reading strategy

- Basics 6, 7, 34, and 35.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Uncoupled two-mode model

Let two local squared frequencies be $a(s) = \omega_1^2(s)$ and $b(s) = \omega_2^2(s)$. Without coupling,

$$\mathbf{H}_0 = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}. \quad (3.1)$$

The eigenvalues cross wherever $a = b$. The corresponding eigenvectors remain pure basis harmonics.

4 Coupled matrix and eigenvalues

Introduce symmetric coupling $g(s)$:

$$\mathbf{H} = \begin{bmatrix} a & g \\ g & b \end{bmatrix}. \quad (4.1)$$

Its squared-frequency eigenvalues are

$$\lambda_{\pm} = \frac{a+b}{2} \pm \sqrt{\left(\frac{a-b}{2}\right)^2 + g^2}. \quad (4.2)$$

For $g \neq 0$, $\lambda_+ \neq \lambda_-$ even at $a = b$. The exact crossing is replaced by an avoided crossing.

5 Gap at the uncoupled crossing

At $a = b = a_c$,

$$\lambda_{\pm} = a_c \pm |g_c|. \quad (5.1)$$

The gap in squared frequency is $2|g_c|$. In ordinary frequency,

$$\Delta\omega = \sqrt{a_c + |g_c|} - \sqrt{a_c - |g_c|}, \quad (5.2)$$

provided $a_c > |g_c|$. For weak coupling,

$$\Delta\omega \simeq \frac{|g_c|}{\sqrt{a_c}}. \quad (5.3)$$

One must distinguish a gap in ω^2 from a gap in ω .

6 Harmonic mixing and mixing angle

Write normalized eigenvectors as

$$\mathbf{v}_+ = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad \mathbf{v}_- = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}. \quad (6.1)$$

The angle satisfies

$$\tan(2\theta) = \frac{2g}{a-b}. \quad (6.2)$$

Far from crossing, each mode resembles one uncoupled harmonic. Near crossing, $\theta \simeq \pi/4$ and both harmonics contribute comparably. Following modes only by ordered eigenvalue therefore makes their apparent identity exchange across the avoided crossing.

7 Positivity-preserving coupling

If $\mathbf{H}$ represents squared frequencies, it should be positive semidefinite in the intended stable local benchmark. For a symmetric 2×2 matrix,

$$a \geq 0, \quad b \geq 0, \quad g^2 \leq ab. \quad (7.1)$$

A useful parameterization is

$$g = \epsilon_c \sqrt{ab}, \quad |\epsilon_c| < 1. \quad (7.2)$$

This automatically respects $g^2 < ab$ where $a, b > 0$ and causes coupling to vanish if either uncoupled branch reaches zero. The current repository uses this form to avoid an unphysical negative local squared frequency.

8 From local matrix to radial differential operator

A global mode has radial structure. One reduced form is

$$-\frac{d}{ds} \left(\mathbf{K}(s) \frac{d\boldsymbol{\xi}}{ds} \right) + \mathbf{H}(s)\boldsymbol{\xi} = \omega^2 \mathbf{M}(s)\boldsymbol{\xi}. \quad (8.1)$$

The diagonal potential terms reconstruct local continuum branches; off-diagonal terms couple harmonics; $\mathbf{K}$ produces radial communication and boundary quantization. A discrete eigenvalue inside a local gap can represent a global Alfvén eigenmode.

9 Mode tracking across scans

Eigenvalues sorted by magnitude can swap character near an avoided crossing. Track a mode using mass-weighted overlap,

$$\mathcal{O}_{ij} = \frac{|\mathbf{x}_i^\dagger \mathbf{M} \mathbf{y}_j|}{\sqrt{(\mathbf{x}_i^\dagger \mathbf{M} \mathbf{x}_i)(\mathbf{y}_j^\dagger \mathbf{M} \mathbf{y}_j)}}. \quad (9.1)$$

Maximum overlap, harmonic content, and radial localization together provide more reliable identity than eigenvalue index alone.

10 Physical meaning in toroidal plasmas

Toroidicity, ellipticity, and helical geometry introduce equilibrium Fourier coefficients that multiply perturbation harmonics and generate off-diagonal coupling. The two-mode model is not a complete stellarator spectrum, but it captures the universal mechanism: symmetry-breaking geometry couples

branches, opens gaps, and permits discrete mixed-harmonic modes.

11 Computational laboratory

Scan a two-by-two matrix through an uncoupled crossing, plot eigenvalues and harmonic weights, and verify the analytic formulas. Compare constant coupling with positivity-preserving coupling. Then solve the repository's global two-harmonic radial benchmark and track modes under changes in coupling strength.

12 Summary

- Geometry-induced coupling converts exact branch crossings into avoided crossings.
- The coupled eigenvalues and harmonic mixing follow analytically from a two-mode matrix.
- Squared-frequency models require positivity-aware coupling choices.
- Radial differential terms convert local gaps into global eigenmode problems.
- Overlap and harmonic content are essential for tracking modes across parameter changes.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
a, b	uncoupled squared frequencies	Diagonal two-mode entries.
g	coupling coefficient	Off-diagonal harmonic interaction.
$\lambda_{\pm}$	coupled squared frequencies	Eigenvalues of the local matrix.
θ	mixing angle	Harmonic composition parameter.
ϵ_c	dimensionless coupling strength	In $g = \epsilon_c \sqrt{ab}$.
$\mathcal{O}_{ij}$	mode overlap	Mass-weighted eigenvector similarity.

13 Exercises

1. Derive the two coupled eigenvalues from the characteristic polynomial.

2. Derive the mixing-angle formula.
3. Convert a gap in squared frequency to a gap in frequency.
4. Prove the positive-semidefinite condition $g^2 \leq ab$.
5. Explain why eigenvalue sorting alone fails as a mode-tracking method.

14 References

References

- [1] G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press (2013).
- [2] G. Strang, *Linear Algebra and Its Applications*, 4th ed., Brooks/Cole (2006).
- [3] W. A. Strauss, *Partial Differential Equations: An Introduction*, 2nd ed., Wiley (2007).
- [4] J. P. Freidberg, *Ideal MHD*, Cambridge University Press (2014).
- [5] J. P. Goedbloed, R. Keppens, and S. Poedts, *Advanced Magnetohydrodynamics*, Cambridge University Press (2010).
- [6] H. Goedbloed and S. Poedts, *Principles of Magnetohydrodynamics*, Cambridge University Press (2004).
- [7] T. H. Stix, *Waves in Plasmas*, American Institute of Physics (1992).
- [8] D. A. Gurnett and A. Bhattacharjee, *Introduction to Plasma Physics*, 2nd ed., Cambridge University Press (2017).
- [9] F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion*, 3rd ed., Springer (2016).
- [10] R. J. Goldston and P. H. Rutherford, *Introduction to Plasma Physics*, CRC Press (1995).
- [11] P. M. Bellan, *Fundamentals of Plasma Physics*, Cambridge University Press (2006).